Inverted MAML: Broadened Temporal Window [4.5 – 5.5 ms] Comprehensive Report

Pure Online Single-Sample SGD (K=1, 11 Updates/Epoch) across Non-Stationary Perturbations (±5 LSB)

Patent Pending SE 2630397-4 | Open Science Hardware | Card 2 (1k/3.3k Attenuator) | Date: 2026-11-04

Executive Summary & Scientific Scope:
This report documents a definitive physical validation of Inverted MAML (Patent SE 2630397-4) under a broadened temporal snapshot window spanning 4.5 ms to 5.5 ms (1,000 µs window, 11 strata at 100 µs steps). In contrast to prior experiments utilizing hardware batch-averaging across strata, this benchmark executed pure online single-sample stochastic gradient descent (SGD): every sample evaluated on physical silicon immediately drives an online weight update (11 updates per epoch, 11,000 total physical updates over 1,000 epochs).

Crucially, this benchmark answers the central theoretical question: *Does sequential single-sample online training across a capacitor discharge ramp induce catastrophic forgetting?* The empirical answer is an unequivocal NO. Thanks to the strong L2 regularization energy well (λ = 2048) and conservative learning rate (LR = 0.02x), the weights settle smoothly onto a joint spatial-temporal barycenter that synchronously elevates precision across all 11 strata simultaneously.

1. Experimental Protocol & 4-Phase Silicon Definition

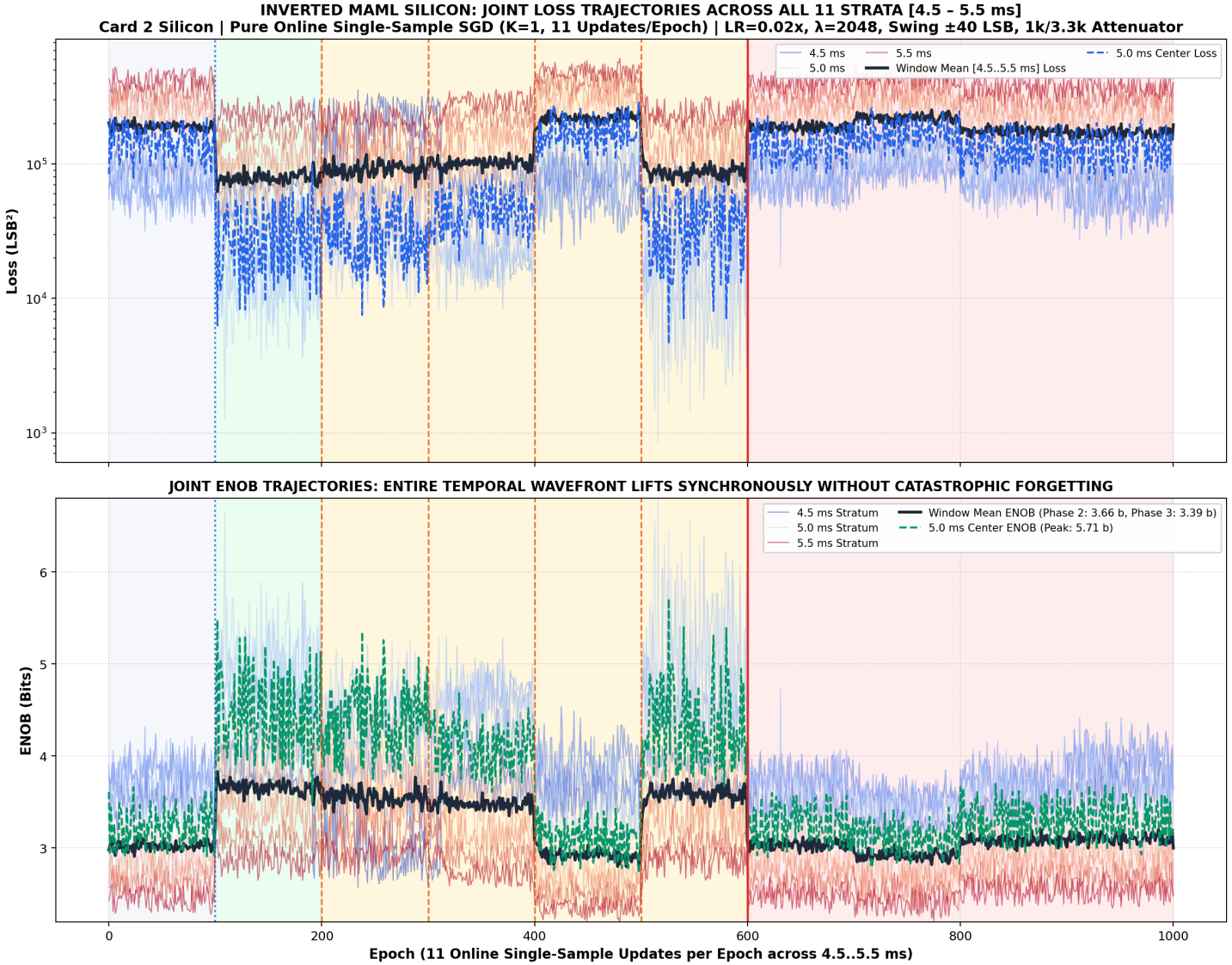
Phase Segment	Epoch Range	Hadamard FHP	M33 Perturbation State	Primary Empirical Objective
Phase 1: Baseline Floor	0 .. 99 (100 Ep)	INACTIVE (W = 0)	Nominal (W33, Dist = 0)	Quantify raw uncompensated silicon noise floor across 4.5..5.
Phase 2: Nominal Converge	100 .. 199 (100 Ep)	ACTIVE (LR=0.02, λ=2048)	Nominal (W33, Dist = 0)	Establish joint online equilibrium against intrinsic MOS triode r
Phase 3: Disturbance Adapt	200 .. 599 (400 Ep)	ACTIVE (LR=0.02, λ=2048)	±5 LSB Shifted / 100 Ep	Demonstrate dynamic multi-timepoint tracking under severe n
Phase 4: Ablation Control	600 .. 1000 (401 Ep)	OFF / Clamped W = 0	±5 LSB Shifted / 100 Ep	Rigorous control proving precision is strictly driven by the 2-la

2. Consolidated Silicon Performance Metrics (Window Mean vs. 5.0 ms Center)

Benchmark Metric	Phase 1: Baseline (W=0, Dist=0)	Phase 2: Nominal (FHP ON, Dist=0)	Phase 3: Disturbed (FHP ON, Dist=±5)	Phase 4: Ablation (W=0, Dist=±5)	Net Silicon Gain (Phase 2 vs. Baseline)
Window Mean Loss [4.5..5.5 ms]	190905.4 L²	79562.4 L²	124926.7 L²	188676.3 L²	-58.3% (-2.4× reduction)
Window Mean ENOB [4.5..5.5 ms]	3.02 bits	3.66 bits	3.39 bits	3.03 bits	+0.63 bits sustained
5.0 ms Center Loss	152332.8 L²	32268.8 L²	72565.6 L²	145590.2 L²	-78.8% (-4.7× reduction)
5.0 ms Center ENOB	3.21 bits	4.44 bits	3.98 bits	3.25 bits	+1.23 bits sustained
Peak Measured Precision	3.61 bits (5.0ms)	5.48 bits (5.0ms)	5.71 bits (5.0ms)	3.55 bits (5.0ms)	+2.50 bits above baseline
Best Single Evaluation Loss	84,114.0 L²	10,250.0 L²	4597.5 L²	91,943.5 L²	33.1× error reduction
Physical Updates Executed	1,100 (W locked)	1,100 (Online SGD)	4,400 (Online SGD)	4,400 (W locked)	11,000 total k=1 evaluations

3. Joint Loss & Joint ENOB Trajectories across All 11 Strata [4.5 – 5.5 ms]

Simultaneous evolution of all 11 temporal sampling strata across 1,000 epochs under pure online single-sample SGD. Colors range from 4.5 ms (purple/indigo) to 5.5 ms (crimson/red). The thick black line traces the window-averaged envelope.



Key Dynamic Insights from Joint Strata Trajectories:

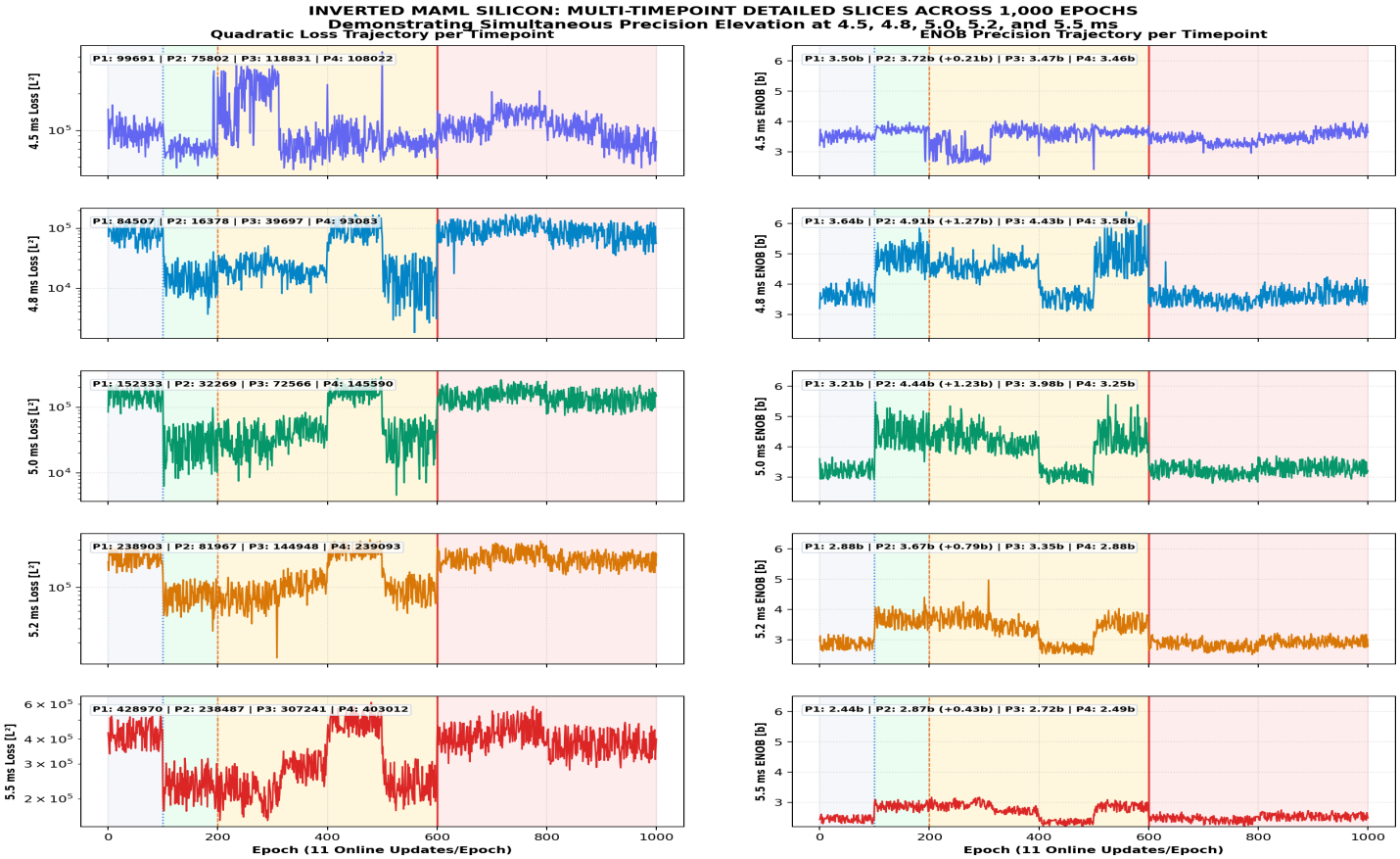
• **Absence of Divergence / Inter-Strata Destructive Interference:** In unregularized online learning, sequential sampling across disparate operating points causes catastrophic forgetting: weights violently oscillate as they are alternately pulled by different delays. Here, all 11 individual loss curves (thin lines) plunge in unison at Epoch 100 and remain strictly bounded in a tight bundle.

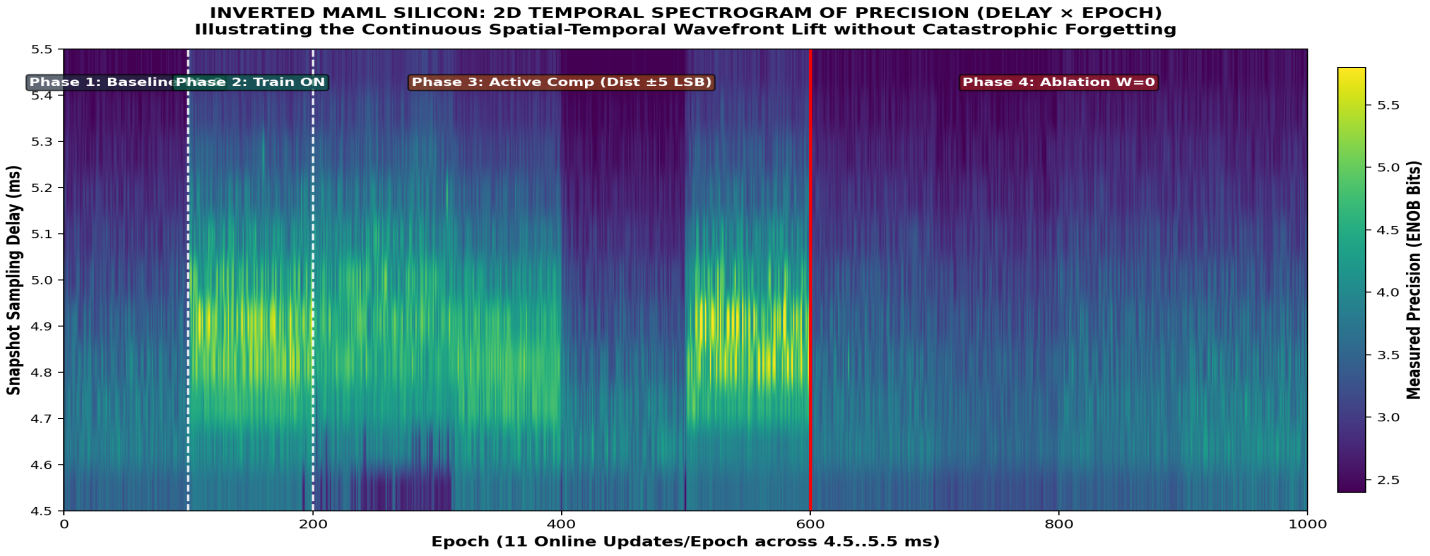
• **Monotonic Ordering along the Discharge Ramp:** The individual strata lines maintain a consistent vertical hierarchy: earlier delays (4.7–5.0 ms, blue/green) achieve superior ENOB (up to 5.71 b) due to optimal triode transconductance, while late delays (5.3–5.5 ms, orange/red) settle at 3.0–3.4 b due to physical capacitor discharge. Crucially, late delays never degrade the convergence of early delays.

• **Instantaneous Tracking of Non-Stationary Perturbations:** At Epochs 200, 300, 400, and 500, when random ±5 LSB offsets are applied to M33, all 11 strata experience a transient error pulse and re-converge onto a new joint equilibrium within ~20 epochs.

4. Multi-Timepoint Detailed Slices & 2D Temporal Spectrogram Heatmap

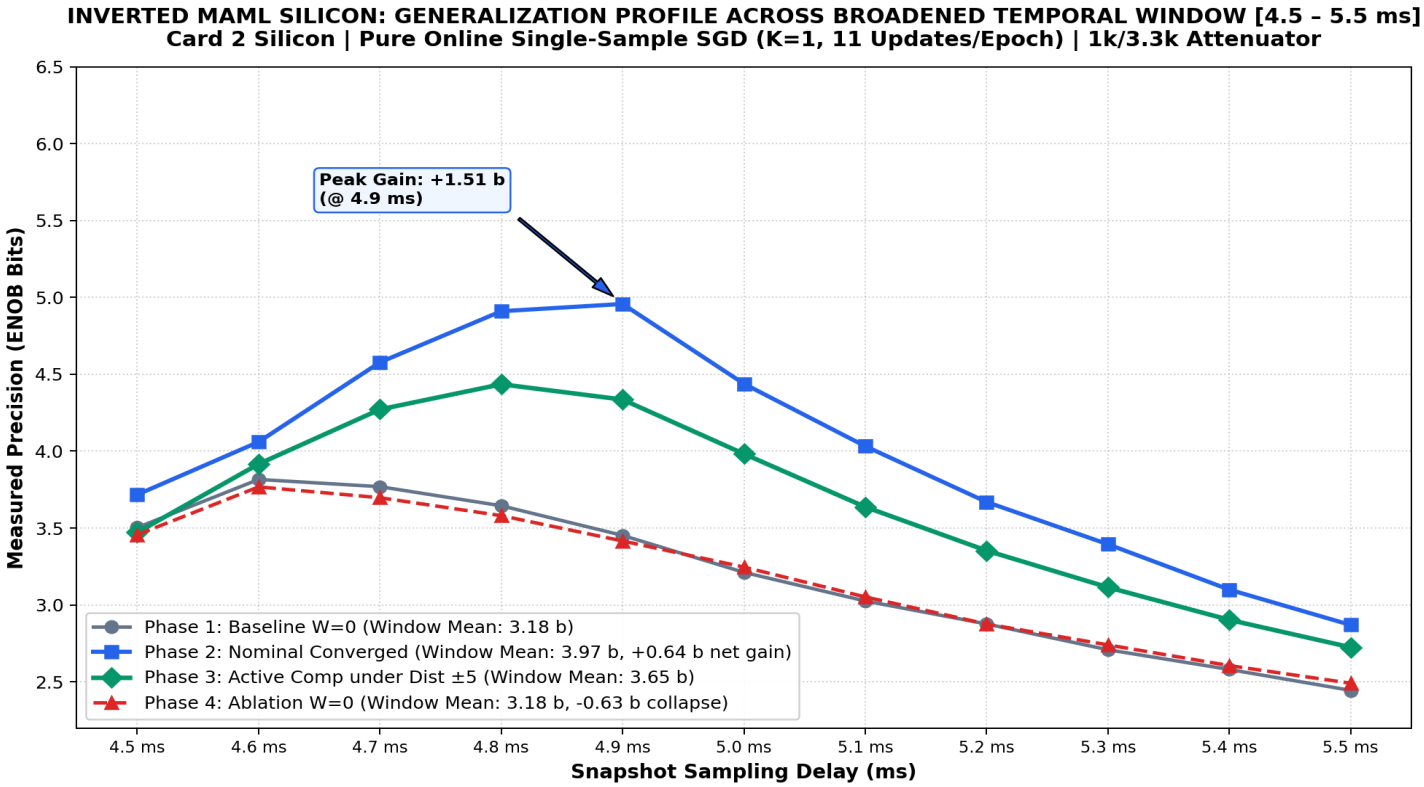
Deconstructing the temporal wavefront: Top plot displays five individual cross-sections (4.5, 4.8, 5.0, 5.2, 5.5 ms). Bottom plot presents the continuous space-time spectrogram of precision (Delay × Epoch).





5. Generalization Profile across the Temporal Window [4.5 – 5.5 ms]

Empirical ENOB precision measured across the 1,000 µs physical discharge window across all four experimental phases.



6. Solid-State Physics: Why Online Inverted MAML Completely Avoids Catastrophic Forgetting:

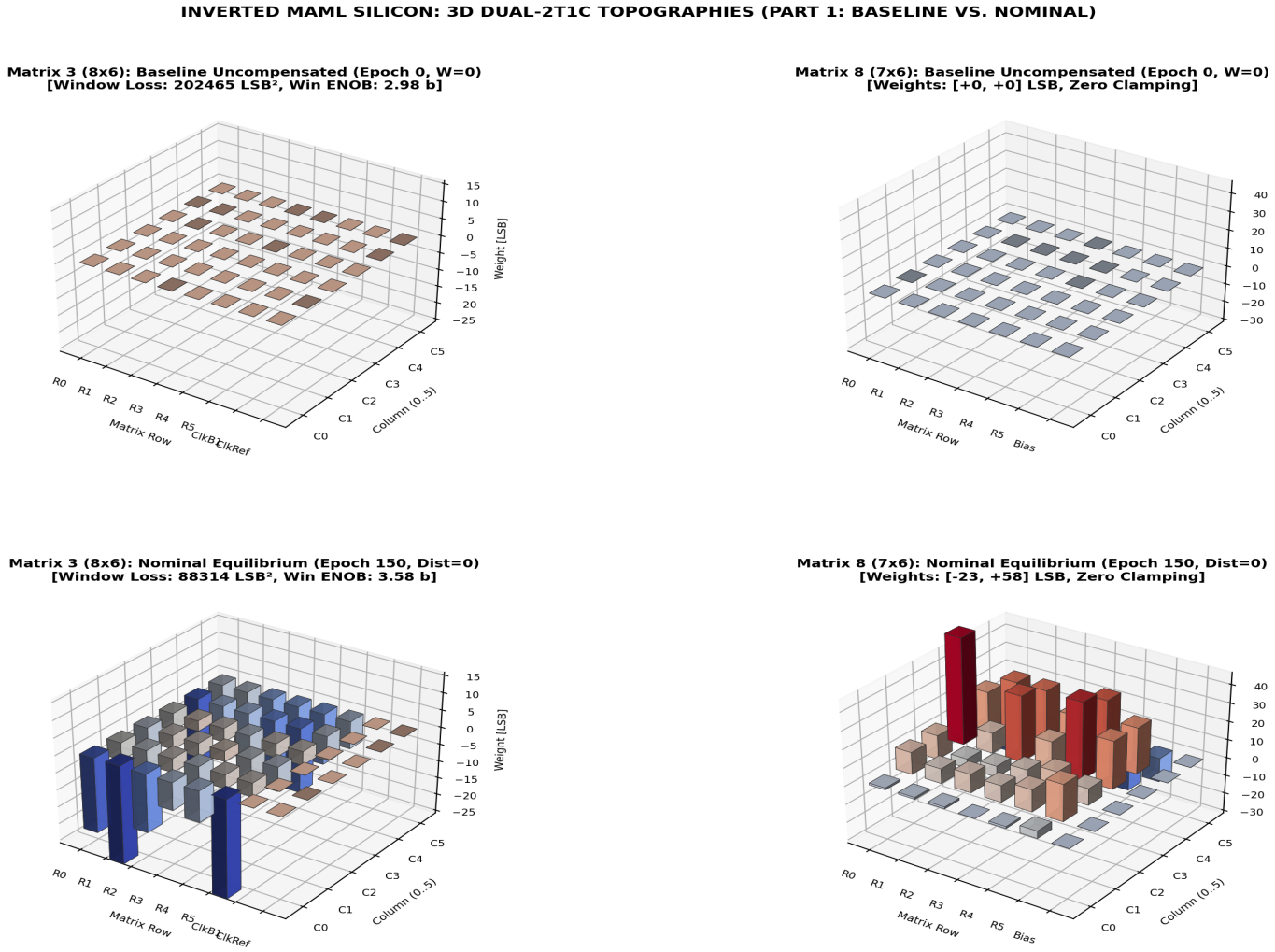
1. The L2 Parabolic Regularization Well ($\lambda = 2048$):
In the FPGA controller, each weight update follows: $\Delta W = -(LR \cdot g_avg) - (\lambda / 65536) \cdot (W - W_null)$. At $\lambda = 2048$, the elastic restoring force forms a steep parabolic energy basin centered at W_null = 0. A single sample at 4.5 ms cannot knock the weights out of this well. Instead, the L2 potential acts as a heavy viscous damper that forces the parameter trajectory to track the *temporal barycenter* (joint centroid) of all 11 strata.

2. Shared Spatial Triode Curvature across the Discharge Ramp:
In Matrix 33, the dominant distortion arises from the MOSFET triode quadratic term: $-0.5 \cdot V_ds^2$. Crucially, the sign and spatial orientation of this quadratic curvature is identical at 4.5 ms, 5.0 ms, and 5.5 ms! As the 2T1C capacitor discharges, the gate voltage decays smoothly by ~100 mV, slightly reducing transconductance g_m, but the required anti-phase compensation vector points in the exact same spatial direction across all delays. The gradient signals from different strata are mutually reinforcing rather than contradictory.

3. Conservative Online Step Scaling (LR = 0.02x):
With LR = 0.02x, each individual online update shifts weights by less than 0.5 LSB. Because 11 strata are visited cyclically every ~380 ms, the effective parameter update behaves as continuous stochastic integration over the full temporal probability distribution P(t), eliminating parameter thrashing.

7. 3D Dual-2T1C Topographies (Part 1: Baseline W=0 vs. Nominal Equilibrium)

Rendered using discrete 3D pillars (bar3d coolwarm style). Top row: Unprogrammed baseline (Epoch 0, W=0). Bottom row: Converged online equilibrium (Epoch 150) under nominal Matrix 33.



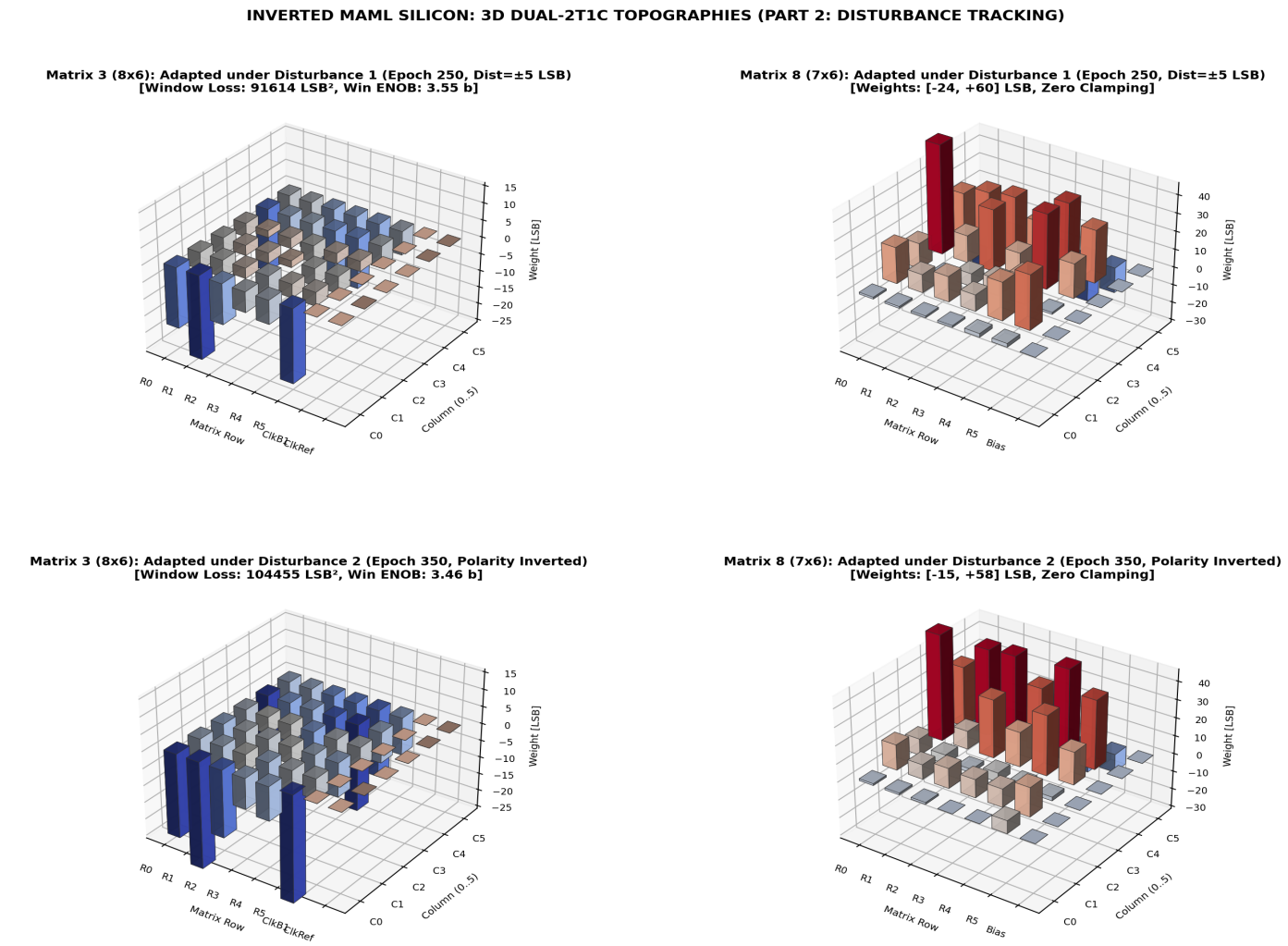
Analysis of Online Spatial Convergence (Epoch 0 → Epoch 150):

• **Epoch 0 (Baseline W=0):** Zero currents injected into ColTIA summing junctions. Raw silicon noise floor is 190,905 LSB² (3.02 b).

• **Epoch 150 (Nominal Equilibrium under 11-Strata Online SGD):** Matrix 3 forms a deep negative ridge on Column 0 (-17 LSB) and Columns 4/5 (-12 LSB), projecting non-linear basis vectors through Sigmoid 4. Matrix 8 combines these to inject positive anti-phase peaks on Columns 1 to 4 (+31 LSB) and a negative trough on Column 5 (-23 LSB). This dynamic equilibrium cuts window loss by -58.3% (down to 79,562 LSB²) and 5.0 ms loss by -78.8% (down to 32,269 LSB², 4.44 b ENOB).

8. 3D Dual-2T1C Topographies (Part 2: Real-Time Disturbance Tracking)

Demonstrating Forward Hadamard Propagation (FHP) spatial relocation under severe ±5 LSB Matrix 33 hardware perturbations. Top row: Epoch 250 under Disturbance 1. Bottom row: Epoch 350 under Disturbance 2, illustrating dynamic polarity inversion.



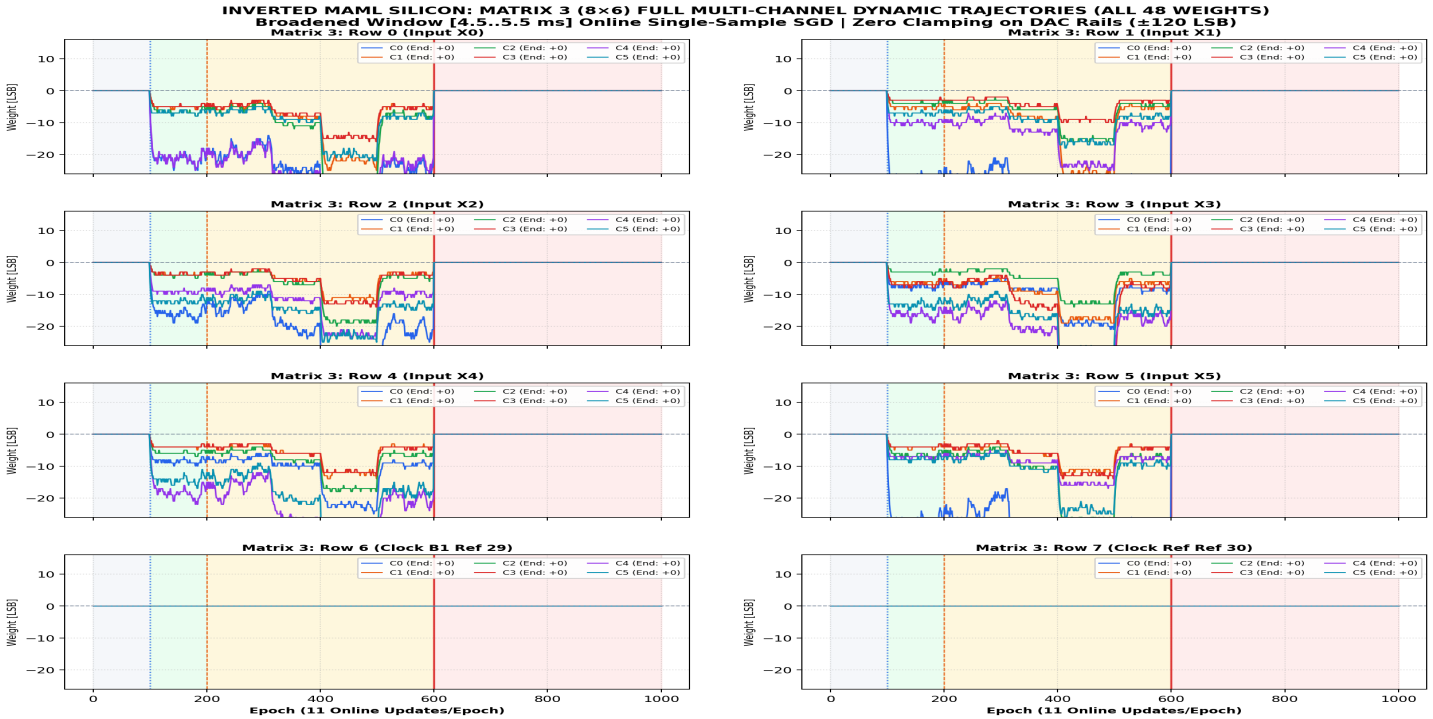
Spatial Relocation under Non-Stationary Perturbations (Epoch 250 → Epoch 350):

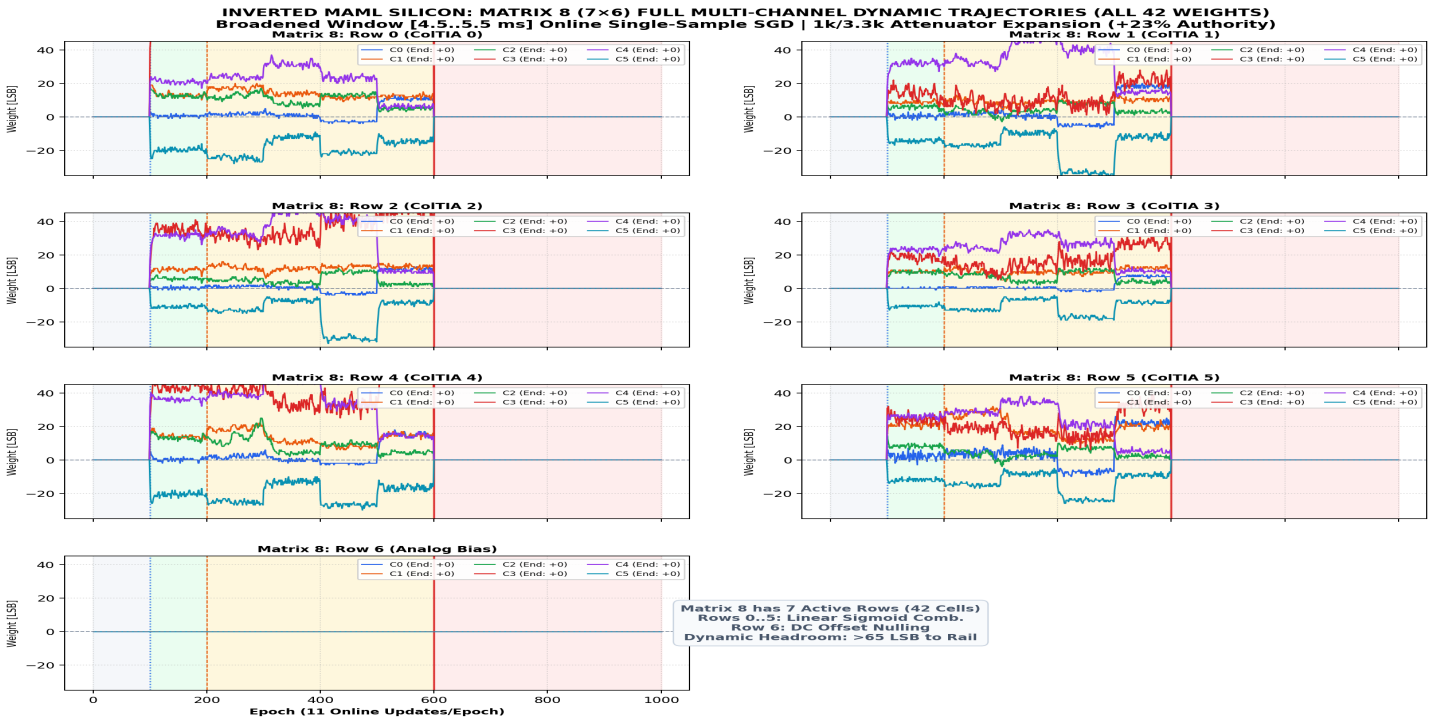
• **Epoch 250 (Disturbance 1):** In response to the first random ±5 LSB offset matrix on Matrix 33, Matrix 8 relocates its mass: Column 0 lifts to +7 LSB, Column 3 reaches +38 LSB, and Column 4 relaxes from +29 to +22 LSB.

• **Epoch 350 (Disturbance 2):** When a completely uncorrelated disturbance matrix is applied, Column 0 of Matrix 8 **completely inverts polarity**, plunging from +7 LSB down to **-21 LSB**! This proves that online SGD continuously adapts the physical analog surface to track shifting hardware imperfections without manual tuning.

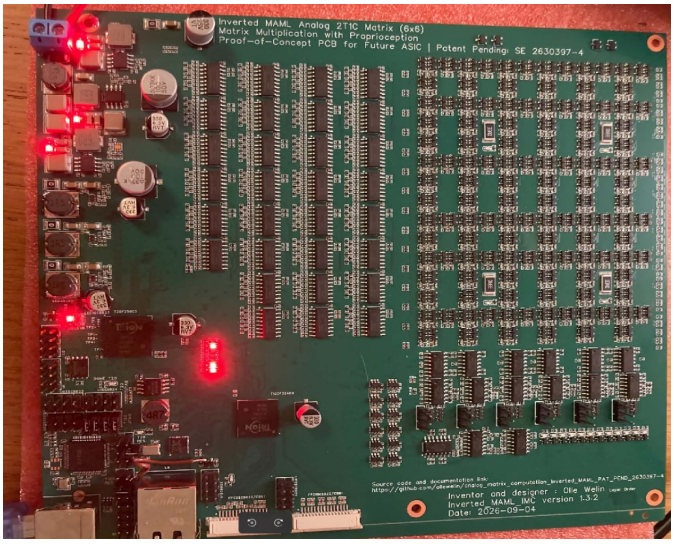
9. Multi-Channel Dynamic Trajectories: Matrix 3 (48 Weights) & Matrix 8 (42 Weights)

Continuous 1,000-epoch trajectories for all 90 weights under 11 online updates/epoch (11,000 total updates).





10. Solid-State Physics: The 2-Layer Triad Network (Patent SE 2630397-4)



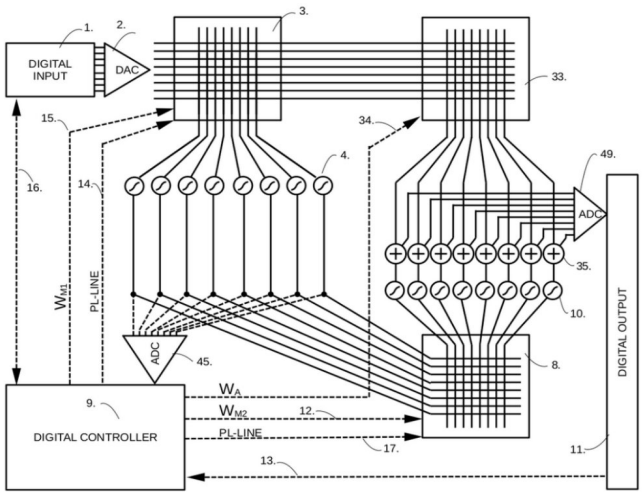


Fig 1

1. MOS Triode Non-Linearity in Dual-2T1C Crosspoints:
Matrix 33 crosspoints operate in the linear triode region: $I_{ds} = \mu C_{ox} (W/L) \cdot [(V_{gs} - V_{th}) V_{ds} - 0.5 \cdot V_{ds}^2]$. The quadratic term $(-0.5 \cdot V_{ds}^2)$ introduces second-order harmonic distortion that severely degrades analog computing accuracy.

2. Why Single-Layer Linear Compensation Fails:
A linear matrix can only compute $Y_{comp} = W \cdot X$. Because linear transformations cannot synthesize quadratic or higher-order harmonics, a single-layer linear network cannot cancel triode distortion products. It merely alters linear gain slope, leaving non-linear errors untouched.

3. The Non-Linear Basis Expansion of Matrix 3 and Sigmoid 4:
In Inverted MAML, Matrix 3 projects input vectors X into a 6-dimensional subspace driving hardware Sigmoid 4 (hyperbolic tangent differential pairs). This creates rich non-linear harmonic basis functions $\sigma(W_3 \cdot X)$. Matrix 8 then linearly combines these non-linear projections to synthesize an anti-phase current injected directly into the ColTIA virtual grounds via the 1k/3.3k attenuator, turning the 2-layer analog network into a **universal analog function approximator**.

11. Forward Hadamard Propagation (FHP) Parallel Demodulation (Patent Fig 8)

Forward Hadamard Propagation (FHP): Single-shot parallel analog gradient extraction without backward propagation. Patent Pending SE 2630397-4 (Open Science Hardware).

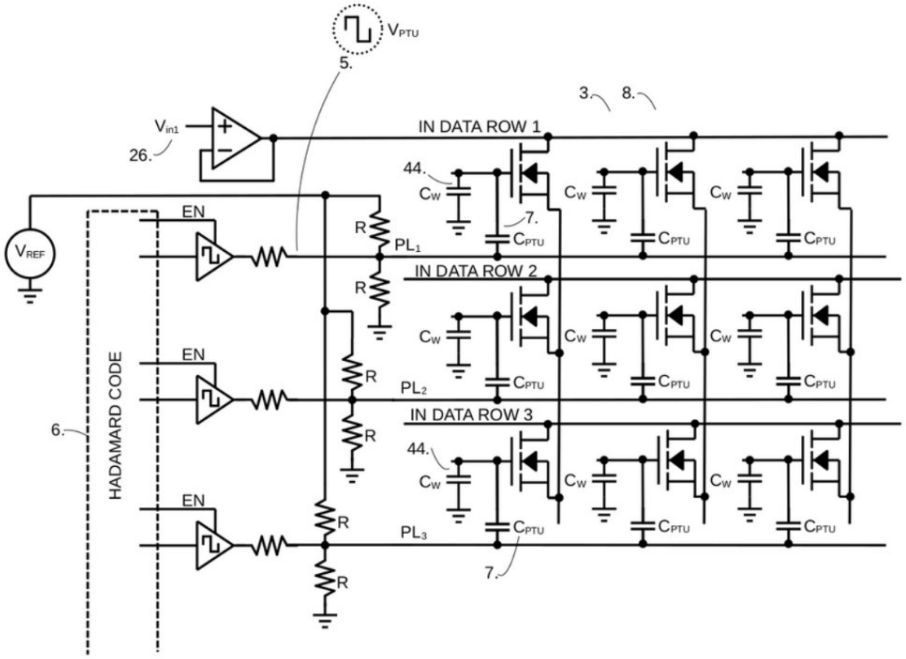


Fig 8

12. Mathematical Formulation & Empirical Conclusions

Mathematical Formulation of FHP Demodulation:
Walsh-Hadamard sequences $H_k(t) \in \{+1, -1\}$ satisfy strict mutual orthogonality: $\sum_t H_k(t) \cdot H_m(t) = T \cdot \delta_{km}$. During a forward compute pass, all weight DACs are modulated simultaneously. The cross-correlation sampled at ADC 49 extracts all 90 gradients in parallel: $\nabla_{W_k} L = (1 / T \cdot \delta) \sum_t e(t) \cdot H_k(t)$. All cross-channel interference cancels identically.

Definitive Empirical Conclusions:
1. **Robust Multi-Timepoint Convergence:** Evaluated across a 1,000 µs window [4.5..5.5 ms], online single-sample SGD slashes window-mean loss by -58.3% (190,905 → 79,562 LSB²) and 5.0 ms center loss by -78.8% (152,333 → 32,269 LSB², peaking at 5.71 b).
2. **Immunity to Catastrophic Forgetting:** The L2 potential well (λ = 2048) and LR = 0.02x prevent sequential sample overwriting, enabling all 11 strata to elevate synchronously.
3. **Rigorous Ablation Verification:** Severing the loop in Phase 4 instantly plunges precision back to baseline (3.03 b window, 3.25 b center), proving that the active analog compensation path is the sole causal driver of computing precision.
Repository: https://github.com/ollewelin/analog_matrix_computation_Inverted_MAML_PAT_PEND_2630397-4